Environmental Sensor Analytics

Live DirectQuery over the Kubernetes pipeline · TimescaleDB read replica

Zone

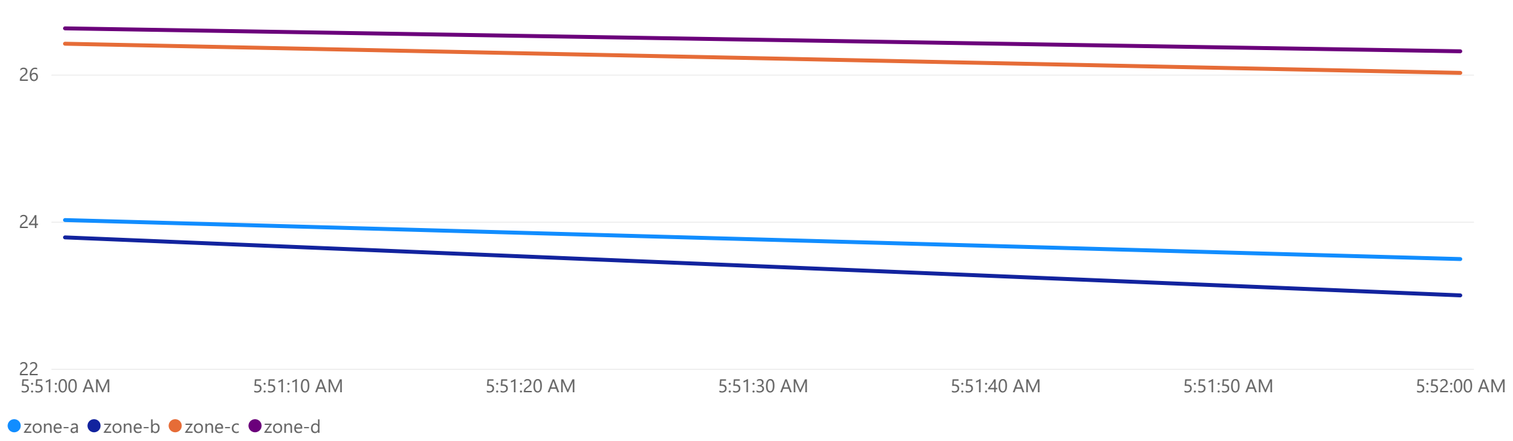
sensor_group

All

120	173.4	80	Moderate	25.0	■ 4 critical	0.002
Sensors Online	Readings per Second	Avg AQI	AQI Category	Avg PM2.5	Alert Status Label	Latency p95 (s)

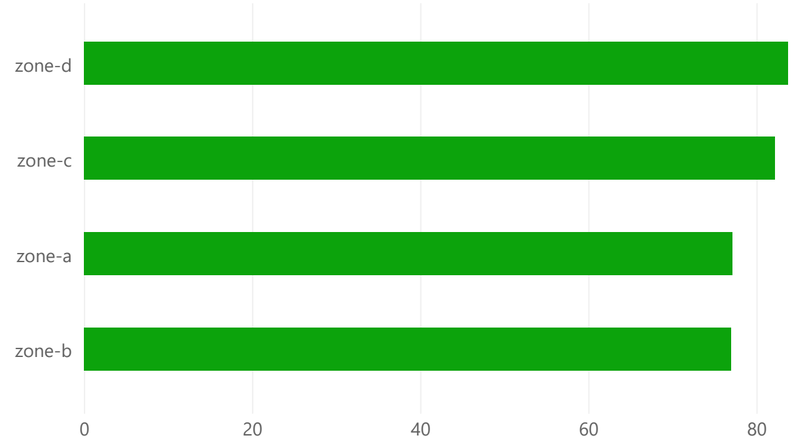
PM2.5 by zone (1-minute average, µg/m³)

Avg PM2.5 by bucket and sensor_group



Average AQI by zone

Avg AQI by sensor_group



Threshold breaches (last 24 h)

bucket	severity	sensor_id	sensor_group	label	value	unit
10/22/2026 5:52:00 AM	warning	node-0008	zone-a	AQI	104.9	
10/22/2026 5:52:00 AM	warning	node-0008	zone-a	PM2.5	37.2	µg/m³
10/22/2026 5:52:00 AM	warning	node-0022	zone-c	AQI	104.1	
10/22/2026 5:52:00 AM	warning	node-0022	zone-c	PM2.5	39.4	µg/m³
10/22/2026 5:52:00 AM	warning	node-0029	zone-b	AQI	145.4	
10/22/2026 5:52:00 AM	warning	node-0029	zone-b	PM2.5	53.6	µg/m³
10/22/2026 5:52:00 AM	warning	node-0031	zone-d	AQI	130.2	
Total					**151.5**	

Sensor drill-down

Pick a zone or a single sensor; every chart and the table follow

Zone

sensor_group

All

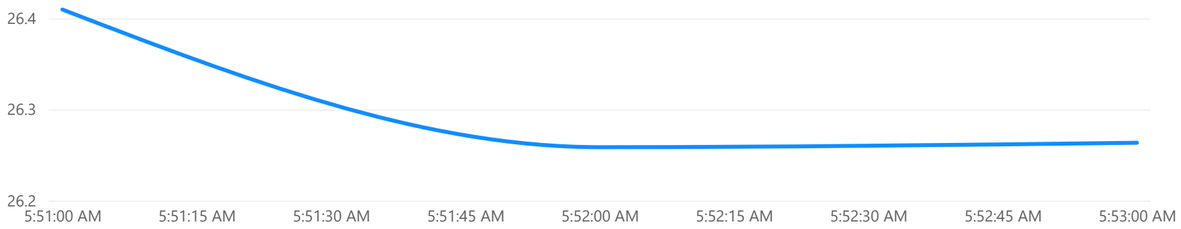
Sensor

sensor_id

All

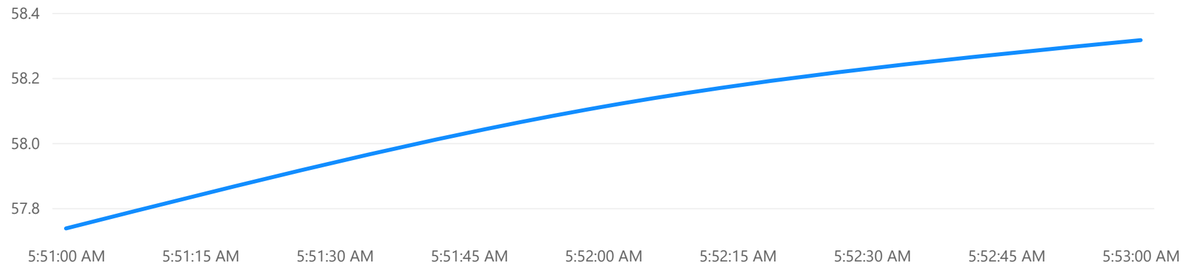
Temperature (°C)

Avg Temperature °C by bucket



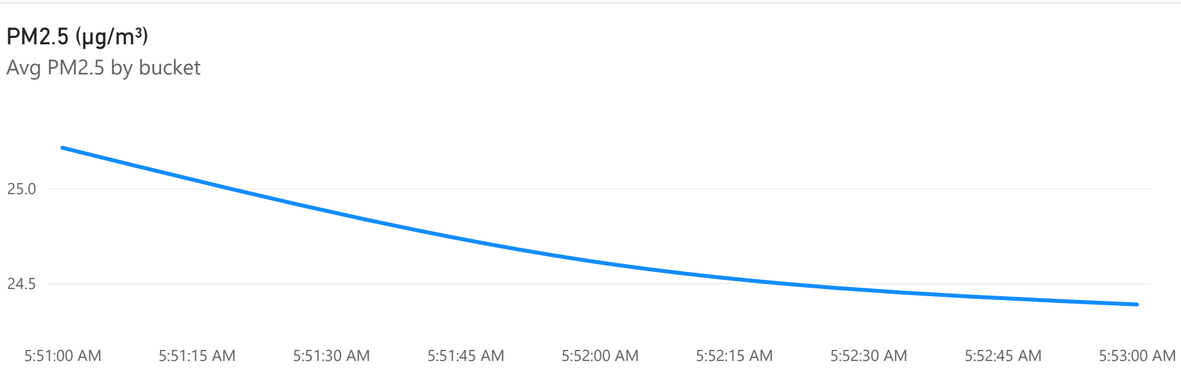
Humidity (%)

Avg Humidity % by bucket



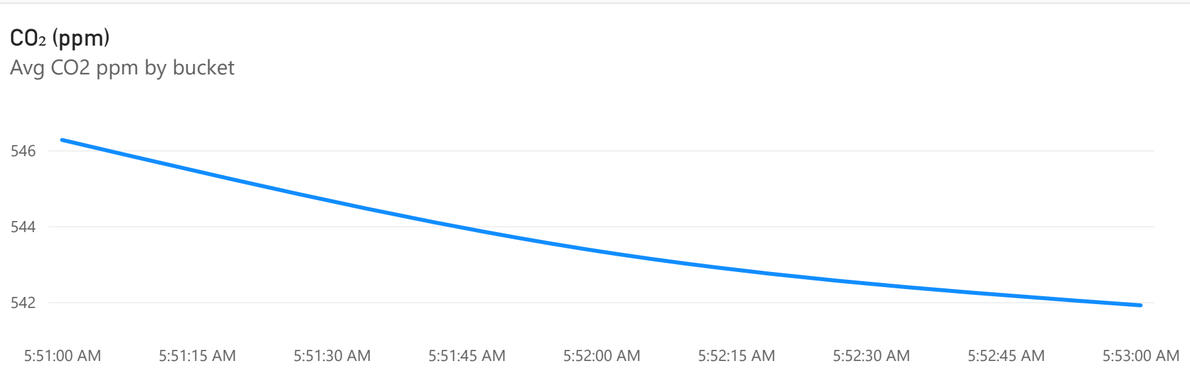
PM2.5 (µg/m³)

Avg PM2.5 by bucket



CO_2 (ppm)

Avg CO2 ppm by bucket



Latest reading per sensor (worst AQI first)

sensor_id	sensor_group	ts	aqi	aqi_category	pm25	pm10	temperature_c	humidity	co2	is_anomaly
node-0107	zone-d	10/22/2026 5:53:02 AM	153	Unhealthy	58.37	93.59	26.13	39.18	575.37	False
node-0029	zone-b	10/22/2026 5:53:02 AM	137	Unhealthy for Sensitive Groups	50.02	85.00	26.19	45.46	662.07	False
node-0031	zone-d	10/22/2026 5:53:02 AM	134	Unhealthy for Sensitive Groups	49.05	91.64	30.74	42.32	728.65	False
node-0073	zone-b	10/22/2026 5:53:02 AM	130	Unhealthy for Sensitive Groups	47.24	76.95	29.33	53.10	468.11	False
node-0062	zone-c	10/22/2026 5:53:02 AM	129	Unhealthy for Sensitive Groups	47.06	84.30	27.12	59.77	588.65	False
node-0095	zone-d	10/22/2026 5:53:02 AM	128	Unhealthy for Sensitive Groups	46.48	82.69	22.49	72.02	552.24	False
node-0119	zone-d	10/22/2026 5:53:02 AM	121	Unhealthy for Sensitive Groups	43.68	78.58	25.39	34.01	726.91	False
node-0034	zone-c	10/22/2026 5:53:02 AM	114	Unhealthy for Sensitive Groups	40.63	68.11	27.18	52.07	549.36	False

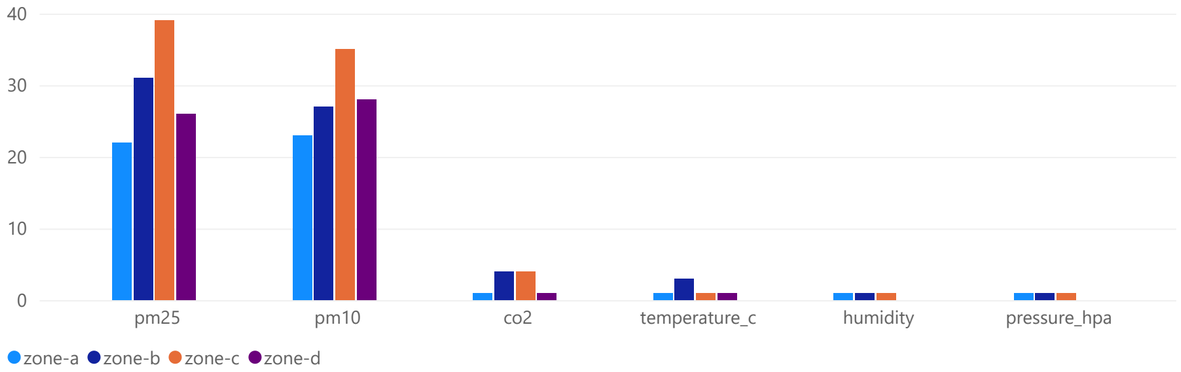
Anomalies and threshold breaches

Anomalies: per-sensor EWMA z-score > 4 · Breaches: 1-minute averages over the thresholds table

144	0.49%	33	7
Anomalous Readings	Anomaly Rate	Sensors in Breach	Critical Breaches

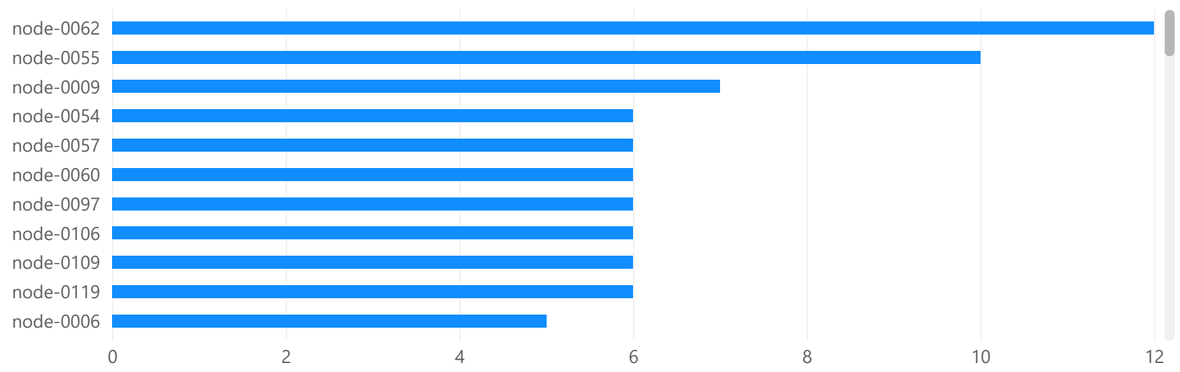
Anomalies by metric and zone (last 24 h)

anomalies by metric and sensor_group



Anomalies per sensor (most first)

anomalies by sensor_id



Anomalous readings

ts	sensor_id	sensor_group	metric	pm25	pm10	aqi
10/22/2026 5:53:05 AM	node-0083	zone-d	pm10	211.0	337.6	286
10/22/2026 5:53:05 AM	node-0083	zone-d	pm25	211.0	337.6	286
10/22/2026 5:53:05 AM	node-0057	zone-b	pm10	107.4	171.9	187
10/22/2026 5:53:05 AM	node-0057	zone-b	pm25	107.4	171.9	187
10/22/2026 5:53:04 AM	node-0082	zone-c	pm10	230.9	369.5	312
10/22/2026 5:53:04 AM	node-0082	zone-c	pm25	230.9	369.5	312
10/22/2026 5:53:03 AM	node-0082	zone-c	pm10	228.8	366.1	308
10/22/2026 5:53:03 AM	node-0082	zone-c	pm25	228.8	366.1	308
10/22/2026 5:53:01 AM	node-0061	zone-b	pm10	245.1	392.2	340
10/22/2026 5:53:01 AM	node-0061	zone-b	pm25	245.1	392.2	340
Total				**604.0**	**966.5**	**500**

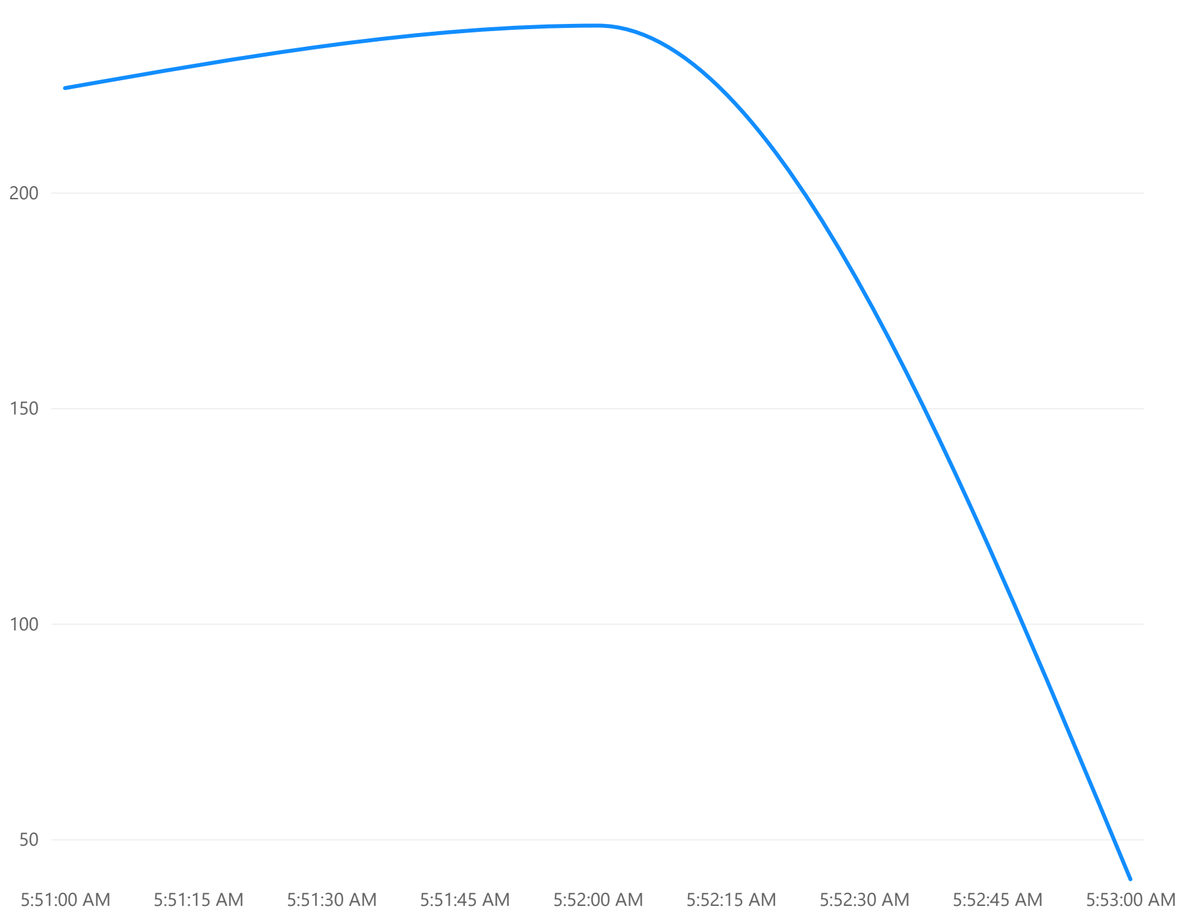
Pipeline health

Throughput and end-to-end latency (sensor timestamp → database commit)

30K	167.7	0.001	0.002	947	149
Readings (last 5 min)	Pipeline Throughput (rec/s)	Latency p50 (s)	Latency p95 (s)	Gap-filled (last 5 min)	Anomalies (last 5 min)

Records committed per second

records per second by minute



End-to-end latency percentiles (seconds)

p50, p95 and p99 by minute

